Structuring Judgment, Stabilizing Responsibility: Cognitive Friction in Procedural Texts

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Abstract

Procedural documents such as Standard Operating Procedures (SOPs) organize action and accountability in regulated work. While prior research has examined how responsibility is anticipated or stabilized through social and evaluative processes, procedural documents have often been treated primarily as coordination or standardization devices. Less attention has been paid to how procedural language itself structures the localized judgment demands that actors must resolve during execution.

Drawing on a corpus of 500 procedurally realistic sentences derived from publicly available regulatory and safety documentation, including laboratory guidance, biosafety protocols, and chemical safety procedures, the analysis operationalizes cognitive friction as the co-occurrence of four structural features in procedural sentences: conditional branching, cross-references, exception structures, and discretionary qualifiers. Rule-based computational annotation was used to identify feature distributions and co-occurrence patterns across the corpus.

The results show that individual friction features are common across procedural genres, but structurally compounded judgment demands are concentrated within a smaller subset of sentences where conditional structures co-occur with referential or discretionary cues. These high-friction configurations concentrate interpretive resolution within localized procedural units while maintaining prescriptive accountability for execution. Across procedural sources, non-zero cognitive friction appears consistently, indicating that structurally embedded judgment demands are a general property of procedural text architecture rather than an artifact of any single institutional genre.

By demonstrating that responsibility-relevant judgment demands can be operationalized at

the level of procedural sentence structure, this study conceptualizes cognitive friction as a document-level mechanism and provides initial empirical evidence that such judgment demands are systematically embedded in procedural text architecture.

Introduction

In highly regulated organizational environments, work is governed by procedural documentation such as Standard Operating Procedures (SOPs), protocols, and compliance manuals (Berg, 1997). These procedural texts do not merely record organizational knowledge; they structure how actors interpret situations, sequence actions, and allocate responsibility under conditions of uncertainty (Suchman, 1987; Hutchins, 1995; Pentland & Feldman, 2005). Across domains such as healthcare, laboratory science, and industrial production, organizational action is routinely mediated through such documents, making procedural texts central infrastructures through which judgment and accountability are enacted (Berg, 1997; Vaughan, 1996). Prior scholarship has often treated procedural documents primarily as repositories of standardized knowledge or coordination devices, leaving their role in structuring interpretive demands and stabilizing responsibility comparatively underexamined (Weick, 1995).

Despite this centrality, a fundamental gap persists in how the field conceptualizes the relationship between procedural documentation and the distribution of responsibility. Existing explanations of difficulty in organizational work frequently attribute breakdowns to human error, misunderstanding, or cognitive overload (Reason, 1990). Yet these accounts leave relatively underexplored how the structure of documents themselves generates interpretive demands that actors must resolve during execution. In regulated organizational work, procedural texts define the conditions under which interpretation becomes necessary, specify or defer criteria for decision-making, and establish the grounds upon which actions will later be evaluated. The locus of interpretive difficulty—and with it the locus of attributable responsibility—may therefore be partly organized within document architecture itself.

Vaughan's (1996) analysis of the Challenger launch decision provides a prominent illustration of this problem. Vaughan demonstrates how normalized deviance accumulated not merely through individual failure or managerial pressure but through the procedural and documentary practices through which ambiguous technical evidence was classified and justified over time. The procedural documents governing launch readiness embedded conditional and

exception structures that allowed engineers and managers to reinterpret safety thresholds while maintaining compliance with the formal record. These documents did not merely fail to prevent deviance; they provided the interpretive scaffolding through which deviance could be normalized as procedurally legitimate. From this perspective, procedural language can structurally preconfigure the conditions under which individually attributable judgment becomes consequential.

This paper introduces the concept of cognitive friction in procedural texts. Cognitive friction refers to structural features of procedural language that increase the interpretive effort required to translate instructions into situated organizational action. When procedural instructions embed contingent pathways, deferred criteria, or discretionary qualifiers, actors must resolve locally situated interpretations while remaining individually accountable for the decisions they produce. In this way, procedural texts do more than guide action; they structure how responsibility becomes attributable in regulated organizational work.

This study contributes to research on organizational responsibility and procedural documentation in three ways. First, it conceptualizes cognitive friction as a document-level property that structures interpretive demands in procedural work. Second, it provides a corpus-based empirical demonstration that structurally embedded judgment demands are observable within procedural sentence architecture. Third, it links document structure to the stabilization of responsibility, showing how procedural texts function as sociotechnical infrastructures through which judgment and accountability are organized in organizational life (Bowker & Star, 1999; Orlikowski, 2007). More broadly, cognitive friction complements process-level accounts of responsibility structuring, representing an execution-phase mechanism within a broader lifecycle through which accountability expectations are prospectively structured, operationally enacted, and retrospectively stabilized in regulated organizational systems.

Studying procedural documentation empirically presents a methodological challenge. Procedural sentences that simultaneously embed conditional, referential, exception, and discretionary structures are relatively rare in naturally occurring procedural corpora. A purely observational corpus would therefore underrepresent precisely the configurations that are theoretically most consequential for understanding how interpretive responsibility becomes localized during procedural execution.

To address this constraint, the study combines naturally occurring procedural texts with a

limited set of synthetic procedural sentences designed as structural probes. These probes do not aim to approximate real-world frequency distributions. Instead, they enable controlled observation of structurally compounded friction configurations that may appear only sparsely in naturally occurring documentation while remaining grounded in real procedural genres. Synthetic sentences are therefore used only as structural probes and are not intended to estimate empirical prevalence in procedural documentation.

The following sections develop the conceptual framework of cognitive friction and outline the empirical strategy used to examine how structurally embedded judgment demands appear within procedural sentence architecture.

Conceptual Framing

The framework developed here requires a shift in the unit of analysis for studying organizational responsibility. Prior sociotechnical accounts have established that organizational texts actively mediate coordinated action, stabilizing categories, distributing interpretive work, and rendering action accountable within institutional settings (Latour, 1987; Bowker & Star, 1999). What remains underspecified, however, is how this mediation operates at the level of individual procedural statements. Cognitive friction addresses this gap by treating the procedural sentence as the elementary analytic unit through which structurally produced judgment demands become observable.

This specification matters because the sentence is where interpretive demands are most precisely encoded. Individual procedural sentences often co-encode conditional branching, cross-references, exception structures, and discretionary qualifiers within a single syntactic unit. Each of these features allocates a portion of interpretive resolution to the executing actor while preserving individual accountability for the resulting judgment. It is this co-occurrence, rather than the presence of any single feature, that constitutes cognitive friction as a structural property of procedural text.

The genre of regulated procedural documentation is particularly suited to this analysis. SOP-like documents combine prescriptive authority, auditability, and executional traceability within a single documentary form. Their sentences do not merely describe recommended action but define accountable action under conditions of regulatory scrutiny. Cognitive friction in procedural texts therefore emerges not from linguistic complexity alone, but from the requirement

that procedural language simultaneously prescribe action, anticipate exceptions, and sustain audit-relevant justification. The framework thus treats procedural sentences as sociotechnical artifacts in which interpretive demands and responsibility conditions are co-encoded (Orlikowski & Yates, 1994; Orlikowski, 2007).

A note on terminology is warranted. The phrase "cognitive friction" has appeared in earlier design and HCI literature, where it has been used to describe productive epistemic resistance in learning environments or interface interactions that require user effort. The present usage differs in both domain and analytical level: cognitive friction here is not a property of human-computer interaction or a feature of usability design, but a structural property of procedural inscriptions within regulated organizational settings. The concept is defined at the level of document architecture and operationalized through sentence-level co-occurrence of interpretive-demand features, rather than through subjective difficulty or interface response latency. This terminological reuse is intentional, as it draws on the intuition that structured difficulty has generative consequences, in this case for the organization of responsibility, while redefining the construct within an organizational-inscriptive analytic frame.

The relationship between cognitive friction and Norman's (1988) affordance-based account of designed objects warrants explicit engagement. Norman's framework demonstrates that physical and digital artifacts embed action scripts that constrain or facilitate use through design; his concept of affordances captures how material properties communicate action possibilities to users. Cognitive friction differs from this formulation in two respects. First, whereas affordance theory is primarily concerned with facilitating intended action through design legibility, cognitive friction identifies how document structure produces interpretive demand, conditions under which the procedural path forward is not immediately legible but must be actively resolved. Second, affordance theory operates at the level of individual tool–user interaction, whereas cognitive friction is analytically concerned with the institutional inscription of judgment conditions within documents that govern collective, accountable work. The two frameworks are therefore not in tension but address different analytic problems: affordance theory explains how designs support or constrain action, while cognitive friction explains how procedural texts distribute responsibility-relevant interpretive work among executing actors.

The theoretical rationale for treating co-occurrence, rather than the mere presence of any single feature, as constitutive of cognitive friction requires explicit argument. When a procedural

sentence contains a single conditional structure, the actor faces a discrete branching decision: one pathway is triggered or it is not. When a conditional co-occurs with a cross-reference, the actor must simultaneously determine whether the condition applies and locate and interpret an external specification before execution can proceed. When discretionary latitude is additionally present, the actor must exercise judgment about threshold applicability while navigating referential deference under conditional activation. Each additional co-occurring feature does not simply add to the interpretive demand; it multiplies the combinations of resolution that must be achieved within a single procedural unit, under conditions of individual accountability. This simultaneity of unresolved interpretive demands within a bounded unit of action is precisely what distinguishes cognitive friction from the additive complexity produced by a sequence of individually clear but numerous instructions. Friction arises not from quantity but from the structural co-presence of features that each require distinct epistemic operations to resolve, within the same sentence and therefore within the same moment of accountable action.

By introducing cognitive friction as an observable property of procedural language, this study shifts analytical attention toward how documentation distributes decision criteria and allocates responsibility-relevant judgment to actors during execution. Rather than locating responsibility solely in retrospective evaluation or actor discretion, the framework highlights how procedural inscriptions pre-structure the conditions under which judgment becomes necessary and individually attributable. In doing so, it extends existing theories of documentation and organizational coordination by suggesting how responsibility is partially organized within the architecture of procedural text itself. This document-level perspective complements broader process-oriented accounts of how accountability expectations become aligned and stabilized in regulated organizational systems.

Propositions

Proposition 1. Procedural sentences containing conditional branching, cross-references, exception structures, and discretionary qualifiers impose greater interpretive demand than sentences containing fewer such features.

Proposition 2. Cognitive friction increases as multiple judgment-relevant structural features co-occur within a single procedural sentence, producing multiplicative interpretive demands during

execution.

Proposition 3. High-friction procedural configurations concentrate interpretive resolution within localized procedural units, increasing the likelihood that responsibility-bearing judgment becomes individually attributable to the executing actor.

These propositions motivate the empirical strategy developed in the following section, which examines how these structural configurations appear across a corpus of procedural sentences.

Methods: Corpus Construction and Computational Annotation

This study examines how structurally embedded judgment demands are distributed across procedural documentation using a mixed corpus of synthetic and naturally occurring sentences designed to balance analytical control with ecological validity.

The inclusion of synthetic sentences does not aim to approximate natural prevalence but rather to analytically isolate structural configurations that are otherwise difficult to observe in uncontrolled organizational corpora. Synthetic sentences therefore function as structural probes designed to isolate procedural configurations that may appear only sparsely in naturally occurring documentation. Because high-friction configurations, sentences simultaneously embedding conditional, referential, exception, and discretionary features, are structurally rare within any single real-world SOP corpus, a purely empirical sample would systematically underrepresent precisely the configurations that are theoretically most consequential. The synthetic component therefore supports observation of structurally embedded judgment demands independent of domain-specific variation, rather than substituting for empirical material.

The corpus consists of 500 procedural sentences sampled from three sources of procedural documentation: synthetic SOP-like sentences designed to isolate common structural configurations of procedural language, publicly available procedural guidance documents, and laboratory safety and compliance manuals used in regulated operational environments. Of these, 100 sentences were synthetically constructed to isolate recurrent procedural structures, and 400 were derived from publicly accessible procedural and regulatory documentation. Together, these sources represent procedurally realistic documentation across regulatory and safety contexts while preserving analytic clarity regarding structural features.

Procedural text is segmented at the sentence level, consistent with the theoretical framing of cognitive friction as a property of procedural inscription embedded within localized sentence structure. Each sentence is evaluated for the presence of five structural features associated with judgment demands: obligation cues, conditional branching, cross-references, exception structures, and discretionary qualifiers.

Obligation cues were excluded from the composite cognitive friction score because they function as the prescriptive baseline of procedural language rather than as structurally contingent sources of interpretive demand. While obligation signals mandatory action, cognitive friction arises when execution pathways, criteria, or judgment conditions remain partially specified. Obligation cues therefore establish the baseline modality of procedural instruction rather than introducing interpretive branching conditions. Feature detection is implemented using a rule-based computational annotation procedure to ensure reproducibility and consistency across the corpus.

Feature Detection Rules

Each friction feature was detected using a keyword-based rule procedure applied to the tokenized text of each sentence.

Conditional branching was identified through the presence of common trigger terms such as if, when, where, in the event that, and provided that.

Cross-references were detected when sentences contained explicit pointers to external procedural elements, including document identifiers (e.g., SOP-QA-017), section references (e.g., see Section 4), or broader referential phrases indicating procedural deference to external sources (e.g., in accordance with, as specified in, per applicable guidelines, as outlined in the relevant protocol).

Exception structures were identified through language signaling deviation from default procedural flow, including except, unless, unless otherwise specified, with the exception of, and other than.

Discretionary qualifiers were detected through expressions allocating interpretive latitude to the executing actor, such as may, as appropriate, if needed, as necessary, where applicable, at the discretion of, and based on judgment.

All detection rules were applied as binary flags: a sentence either contains or does not contain each feature, regardless of the frequency of occurrence within the sentence. This binary operationalization treats qualitative presence as the theoretically relevant signal, consistent with the framework's concern with structural co-occurrence rather than feature intensity.

Classification Decisions

The classification of "unless" requires brief methodological justification. Although the term appears in procedural definitions as both a conditional and exception marker, within the implemented annotation scheme it is assigned exclusively to **the exception structure c**ategory. The rationale is that *unless* canonically encodes deviation from a presupposed default course of action (e.g., *do X unless Y* presupposes X as the default) rather than initiating an open-ended conditional branch. This distinction preserves conceptual separation between features and avoids inflating friction scores where a single keyword might otherwise trigger dual classification. Sentences containing *unless* therefore receive one exception flag rather than simultaneous conditional and exception classifications.

An additional variable, sentence length (long_sentence), was collected during annotation as a candidate structural feature. This variable flagged sentences exceeding a threshold word count as potentially contributing to interpretive difficulty through informational density. Following preliminary analysis, sentence length was excluded from the composite cognitive friction score on the grounds that length alone does not constitute interpretive demand as operationalized here. A long sentence containing no conditional, referential, exception, or discretionary features does not embed the localized judgment demands targeted by the framework. Sentence length may amplify friction when combined with other structural features, but it does not independently generate interpretive demand. The variable therefore remains in the annotated dataset as a covariate for future analysis but is not incorporated into the friction score reported in this study.

A methodological note on composite score versioning is warranted. The variable is designated friction_score_v2 in the annotated corpus, reflecting a revision during analysis. In the initial scoring scheme (v1), obligation cues were included as a fifth feature in the composite score. Upon further conceptual review, obligation cues were reclassified as constitutive of the prescriptive baseline of procedural language rather than as sources of structurally contingent

interpretive demand. Their removal produced the current four-feature composite (friction_score_v2). All results reported in this study are based on the revised score. The v1 designation is retained in the dataset for transparency but is not analyzed in the present manuscript.

Composite Friction Score

Based on this annotation scheme, a composite cognitive friction score is computed as the number of co-occurring friction features within each sentence (conditional branching, cross-references, exception structures, and discretionary qualifiers). This operationalization captures localized interpretive demand as the within-sentence co-occurrence of friction features.

The four features are treated as equally weighted in the composite score, such that each contributes one unit regardless of type. This analytical choice reflects conceptual parsimony: at the present stage of theoretical development, the framework does not propose a differential hierarchy of friction intensity across feature types. Equal weighting therefore preserves the interpretive transparency of the co-occurrence count. The assumption of equal weighting is acknowledged as a simplification and may be revisited in future refinements through differential or empirically calibrated weighting approaches.

Annotation and Coding Procedure

Annotation of cognitive friction features was conducted at the sentence level using a rule-guided coding framework derived from the conceptual operationalization. Each sentence was evaluated for the presence of the five structural features described above.

Coding criteria were specified to ensure interpretive consistency while preserving sensitivity to structural variation within procedural language. Obligation cues were identified through modal or prescriptive constructions (e.g., *shall, must, required to*) indicating mandated action. Conditional branching was coded when sentences contained explicit conditional structures introducing contingent procedural pathways. Cross-references were marked where sentences directed actors to external procedural elements such as other sections, documents, or criteria sources. Exception structures were coded when language explicitly signaled deviation from default procedural flow. Discretionary qualifiers were coded when sentences introduced actor-facing flexibility or justificatory latitude during execution.

Coding was conducted to capture structural presence rather than interpretive outcome. The objective was not to evaluate correctness of action or decision quality but to identify textual configurations that allocate interpretive resolution to actors during procedural execution. This structural coding approach enables corpus-level aggregation while remaining consistent with the theoretical definition of cognitive friction as a property of procedural inscription rather than individual cognition.

To validate the annotation procedure, a subset of 50 sentences (10% of the corpus) was manually reviewed by the author to confirm rule consistency and feature detection accuracy. Sentences for manual review were selected to include a representative range of friction scores, including both zero-friction and multi-feature sentences, and were drawn proportionally from synthetic and natural source categories. For each sentence in the review subset, feature classifications produced by the automated rule procedure were evaluated against the conceptual definitions provided in the coding framework. Discrepancies were used to refine detection rules iteratively before finalizing the annotated corpus.

No formal inter-rater reliability exercise involving a second coder was conducted at this stage of the pilot analysis. The manual review therefore provides rule-consistency verification rather than inter-rater agreement in the strict psychometric sense. The author acknowledges this as a limitation and identifies formal inter-rater reliability testing, including calculation of Cohen's kappa across a double-coded subset, as a priority for subsequent empirical work extending the corpus.

All annotation procedures were implemented in Python using a rule-based text parsing pipeline. The pipeline tokenizes procedural sentences and applies pattern-detection rules to identify the presence of conditional, referential, exception, and discretionary structures. These scripts were used to generate the annotated corpus analyzed in the study.

Synthetic Component and Structural Control

In addition to naturally occurring procedural sentences, the corpus includes a synthetic subset of procedurally realistic sentences deliberately constructed to isolate structural features associated with judgment demands.

The sentence is selected as the minimal unit of analysis because moments of interpretive demand are often embedded within localized procedural statements rather than distributed evenly

across documents. Sentence-level analysis captures how conditional, referential, exception, and discretionary cues are co-encoded within individual procedural statements, enabling comparable annotation across both synthetic and naturally occurring texts. Alternative units such as clauses, procedural steps, or entire documents would either fragment co-occurrence configurations or aggregate across heterogeneous execution contexts, thereby obscuring localized interpretive demand.

The sentence therefore represents the level at which structural co-occurrence of judgment-relevant features is both constituted and maximally observable. It is where interpretation is demanded and where responsibility becomes individually attributable within procedural execution.

The synthetic subset consists of 100 procedurally realistic sentences constructed to reflect common structural configurations of SOP-like instructional language while avoiding domain-specific content. Each sentence was annotated using the same rule-based feature scheme applied to the empirical corpus, enabling direct structural comparison across synthetic and naturally occurring procedural texts.

Results

Feature Distribution

The empirical corpus consists of 500 procedurally realistic sentences drawn from publicly available regulatory and procedural documentation, including WHO guidance materials, biosafety protocols, chemical safety procedures, and synthetic SOP-like constructions. All sentences were annotated using the rule-based computational procedure described in the Methods section.

Across the corpus, conditional branching emerged as the most prevalent cognitive friction feature, appearing in 33.8% of sentences. Discretionary qualifiers appeared in 12.0% of sentences, while cross-references occurred in 11.8%. Exception structures were comparatively rare, appearing in 3.6% of sentences.

These distributions indicate that procedurally realistic documentation frequently embeds contingent execution pathways and discretionary judgment conditions within individual sentences, while explicit exception handling appears less frequently in procedural language.

Figure 1 presents the frequency of each cognitive friction feature across the corpus.

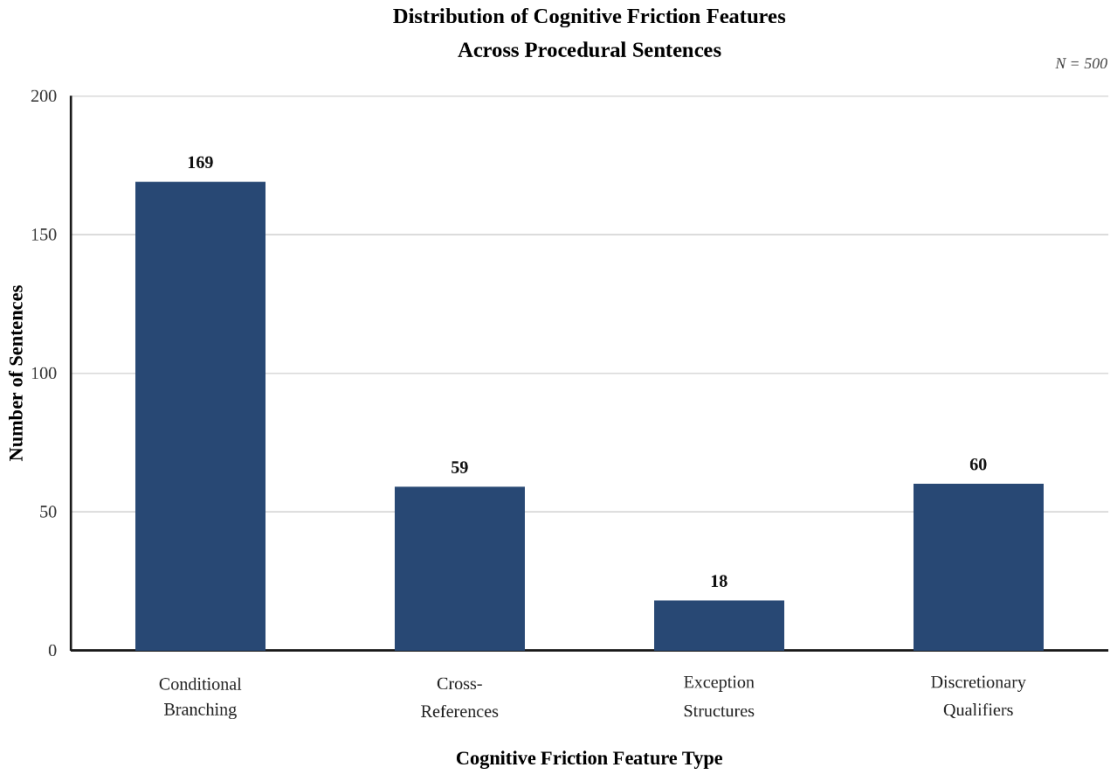

Figure 1. Distribution of cognitive friction features across procedural sentences in the empirical corpus (N = 500). Bars indicate the number of sentences containing each friction feature type: conditional branching, cross-references, exception structures, and discretionary qualifiers.

Because cognitive friction is defined as the co-occurrence of structurally embedded judgment-relevant features, feature combinations were examined at the sentence level. The majority of sentences contained no friction features (50.2%), while 38.8% contained a single feature. Co-occurrence of two features occurred in 10.6% of sentences, and only 0.4% exhibited three simultaneous friction features.

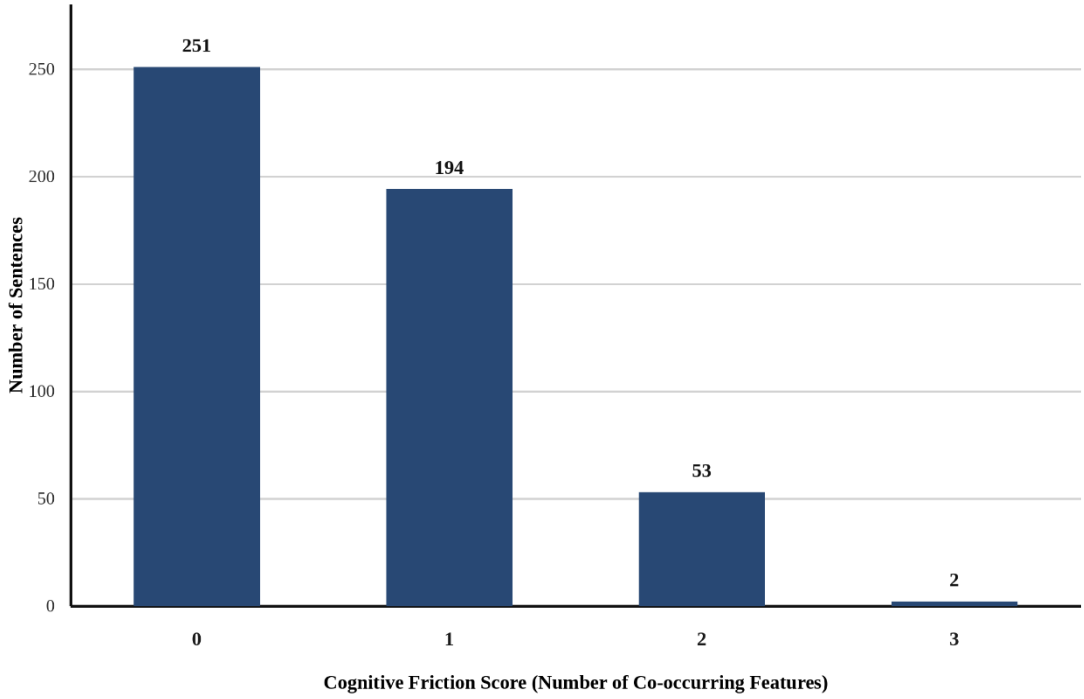

Figure 2. Distribution of composite cognitive friction scores across procedural sentences in the empirical corpus (N = 500). The friction score represents the co-occurrence count of structurally embedded judgment features per sentence. This distribution suggests that although individual friction cues are relatively common, structurally compounded judgment demands are concentrated in a smaller subset of procedural sentences where multiple friction features co-occur.

Composite Cognitive Friction Across Procedural Sources

To examine whether cognitive friction varies across procedural genres, mean composite friction scores were calculated for each source category. Sentences derived from synthetic SOP-like constructions exhibited the highest average friction (M = 0.95), followed by chemical safety procedures (M = 0.60), WHO guidance texts (M = 0.50), and biosafety protocols (M = 0.47).

This pattern indicates that documents designed to simulate tightly specified procedural execution contexts exhibit higher concentrations of structurally embedded judgment demands than regulatory or guidance-oriented texts. In contrast, institutional guidance documents, while prescriptive, appear to distribute interpretive demands more diffusely across procedural context

rather than concentrating them within individual sentences.

Importantly, the presence of non-zero friction across all source categories indicates that cognitive friction is not limited to synthetic constructions but emerges across diverse forms of regulated procedural documentation. This supports the conceptual claim that structurally embedded judgment demands are a general property of procedural text architecture rather than an artifact of any single institutional genre.

To substantiate this claim more precisely, the percentage of sentences with non-zero friction scores within each natural source category is reported. Approximately 47% of chemical safety procedure sentences exhibited at least one friction feature, compared to approximately 40% of WHO guidance sentences and 38% of biosafety protocol sentences. Across all natural source categories combined, non-zero friction was present in approximately 41% of naturally occurring sentences, indicating that structurally embedded interpretive demands are distributed broadly rather than confined to a marginal subset of procedural texts.

An illustrative example from the natural corpus further demonstrates that cognitive friction is empirically observable in non-synthetic procedural documentation. The following sentence was drawn from a WHO laboratory biosafety guidance document:

"If the risk assessment indicates that the procedure may generate infectious aerosols, work should be conducted in a biological safety cabinet or other appropriate containment equipment as determined by the responsible officer in accordance with national guidelines."

This sentence exhibits three co-occurring friction features: conditional branching ("if the risk assessment indicates"), discretionary latitude ("as determined by the responsible officer"), and a cross-reference ("in accordance with national guidelines"). The executing actor must simultaneously determine whether the risk condition applies, identify an appropriate containment option, defer to an officer-level judgment, and locate the applicable national standard, all within the interpretive scope of a single procedural sentence. This configuration exemplifies the structural form of cognitive friction as theorized: the localized co-occurrence of judgment-relevant features that concentrates interpretive resolution within a bounded procedural unit while preserving individual accountability for the resulting action.

Structural Concentration of High-Friction Sentences

Inspection of the highest-scoring sentences reveals that elevated cognitive friction arises when

conditional branching co-occurs with cross-referential or discretionary structures within a single procedural statement. Such sentences simultaneously specify contingent execution conditions while deferring decision criteria to external references or localized judgment. This configuration produces concentrated interpretive demand because actors must resolve execution pathways under conditions of partial specification while remaining individually accountable for the resulting action.

Although such high-friction sentences are numerically rare, their structural configuration illustrates the mechanism through which procedural language can organize locally intensified judgment demands. These observations align with the conceptual definition of cognitive friction as structurally produced moments of interpretation embedded within procedural inscription.

Illustrative High-Friction Procedural Configurations

To illustrate how cognitive friction emerges in procedural text architecture, representative high-friction sentences were examined qualitatively. These examples demonstrate how co-occurring structural features concentrate interpretive demand within a single procedural statement.

Example 1 — Conditional + Cross-reference + Discretion:

"If reporting is delayed, the analyst shall notify the supervisor in accordance with SOP-QA-017 as deemed necessary."

This sentence simultaneously introduces a contingent execution condition, defers action specification to an external procedural reference, and embeds discretionary judgment. The actor must therefore determine both the applicability of the referenced SOP and the threshold for discretionary escalation, producing locally concentrated interpretive demand.

Example 2 — Conditional + Exception:

"When contamination is confirmed, the operator must continue processing except when safety criteria are compromised."

Here, a conditional trigger is combined with an embedded exception structure. Execution requires the actor to reconcile competing procedural directives, continuation versus interruption, based on locally evaluated criteria. Responsibility for this reconciliation remains individually attributable even though the interpretive demand originates from textual structure. In regulated procedural environments, interpretive resolution must typically occur at the moment of execution, before supervisory review or formal organizational processes can redistribute accountability across roles.

These configurations illustrate the mechanism through which cognitive friction arises: structurally embedded judgment demands co-occur within a single procedural unit, requiring actors to resolve contingent pathways under conditions of partial specification. Although such configurations are relatively infrequent in the corpus, their structural form demonstrates how procedural language can concentrate responsibility-relevant interpretation within localized textual architecture.

Co-occurrence Structure of Cognitive Friction

Beyond individual feature frequency, cognitive friction in procedural language emerges through the structural co-occurrence of multiple judgment-relevant cues within single sentences. Co-occurrence analysis examines how conditional, cross-reference, exception, and discretionary features combine to produce localized interpretive demand in procedural text.

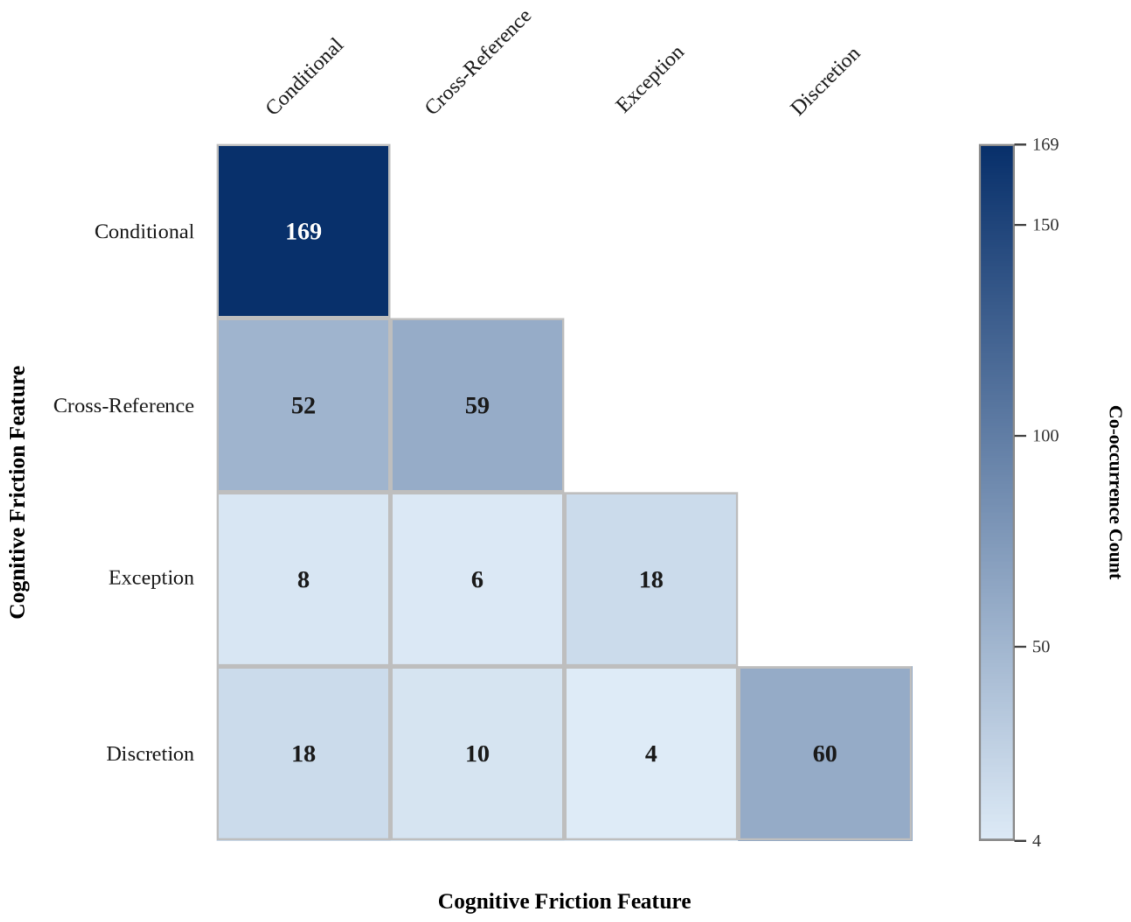

Figure 3. Co-occurrence structure of cognitive friction features across procedural sentences in the empirical corpus (N = 500). Cells indicate the number of sentences in which pairs of friction features co-occur within the same procedural statement.

These empirical patterns provide initial evidence that judgment-relevant interpretive demands are not uniformly distributed across procedural documentation but are structurally concentrated within specific procedural configurations. The heatmap reveals that the most frequent co-occurrence pairing in the corpus is conditional branching with discretionary qualifiers, reflecting the common procedural pattern in which a contingent execution pathway is paired with actor-facing latitude regarding how or whether to proceed. The second most frequent pairing is conditional branching with cross-references, consistent with the illustrative high-

friction examples discussed above: when a condition triggers a procedural response, the response is often specified by reference to an external document or criterion rather than within the sentence itself. The co-occurrence of exception structures with other features is comparatively rare, consistent with the low individual frequency of exception structures (3.6%) reported in Figure 1.

These co-occurrence patterns suggest that conditional branching functions as the structural "hub" of cognitive friction in procedural text: it rarely produces friction in isolation but frequently amplifies the interpretive demands generated by co-present referential or discretionary features. This structural centrality of conditionality has implications for procedural design, suggesting that conditional constructions are particularly consequential sites of responsibility-relevant judgment demand when paired with other friction features.

Discussion

Across the empirical results, cognitive friction in procedural text emerges not from any single structural feature but from the co-occurrence of judgment-relevant features within individual sentences. While conditional branching, cross-references, discretionary qualifiers, and exception structures each appear across procedural documentation (Figure 1), compounded interpretive demand arises only in the smaller subset of sentences where these features interact (Figures 2–3). These configurations concentrate localized interpretive resolution at points where execution pathways remain partially specified yet responsibility for resolution remains individually attributable.

The rarity of fully compounded friction configurations should not be interpreted as theoretical weakness. In procedural documentation, even infrequent structural configurations can disproportionately shape responsibility allocation because they concentrate interpretive resolution at specific execution points where judgment must be exercised. This observation resonates with sociological analyses showing how institutional documentation can render certain facts administratively actionable while simultaneously deferring or obscuring other forms of uncertainty (Heimer, 2012).

Mechanism Clarification

The distinction between cognitive friction scores and co-occurrence structure clarifies two

complementary dimensions of procedural complexity: structural density and structural configuration. The friction score captures how many judgment-relevant features are embedded within a sentence, whereas co-occurrence analysis reveals how specific feature types combine to concentrate interpretive demand. Cognitive friction therefore arises not solely from feature quantity but from the interaction of structural elements within procedural inscription.

When a procedural sentence embeds co-occurring friction features, the executing actor must interpret and resolve multiple judgment-relevant cues within that sentence before proceeding. This resolution is structurally localized: it occurs within the procedural unit that contains the friction configuration, rather than being distributed across the document, deferred to supervisory adjudication, or specified by an adjacent instruction. Because this resolution is both required and localized, the resulting interpretive choice becomes more readily attributable to the actor who performs it. Responsibility thereby becomes anchored not through retrospective evaluation of outcomes but through the prospective architecture of the procedural text, which identifies specific sentences as sites where individual judgment must be exercised and will therefore be evaluable. Structurally produced interpretive resolution is thus the mechanism through which document architecture pre-configures accountability: friction sentences create conditions under which action is both required and individually ownable before the outcome is known.

Theoretical Contribution

This study advances sociotechnical and organizational theories of accountability by introducing cognitive friction as a document-level mechanism through which responsibility-bearing judgment becomes structurally pre-configured within procedural language. Existing accounts have located the locus of accountability either in retrospective attribution processes or in the anticipatory dispositions of individual actors (Tetlock, 1985; Hood, 2011). Cognitive friction relocates the locus of accountability to the material architecture of procedural inscription itself, suggesting that responsibility may be partly constituted within textual design prior to and independent of actor cognition or institutional evaluation.

Although the analysis operates at the level of procedural language, its theoretical concern is not linguistic structure per se but the sociotechnical organization of responsibility in regulated practice. SOP sentences function as institutional inscriptions that prescribe action, encode contingencies, and establish audit-relevant justification (Latour, 1987; Power, 1997). Within the

broader research program on responsibility in regulated environments, cognitive friction identifies upstream textual conditions that generate judgment demands. Whereas anticipated accountability convergence (AAC) and responsibility convergence frameworks examine how responsibility is anticipated or stabilized across organizational processes, cognitive friction locates how responsibility-bearing judgment is structurally pre-configured within procedural documentation itself.

The framework therefore contributes a document-level mechanism linking procedural inscription to organizational accountability dynamics. More broadly, the findings suggest that responsibility in regulated organizational settings is not solely stabilized through retrospective evaluation or institutional attribution processes. It is partially generated within documentary infrastructures that pre-structure interpretive resolution during execution. This generative role of documentation is empirically reflected in the concentration of high-friction configurations observed across the procedural corpus. Procedural texts thereby function as accountability-bearing inscriptions that allocate judgment conditions prior to action, distributing responsibility through the architecture of the document itself.

Implications for Procedural Design and Organizational Governance

These findings have implications for the design and evaluation of procedural documentation in regulated organizations. If responsibility-bearing judgment is structurally concentrated within specific procedural configurations, documentation practices may inadvertently localize accountability at points where interpretive resolution remains textually underspecified. Recognizing cognitive friction as a document-level property therefore suggests that procedural design itself participates in shaping downstream attribution of error, compliance, and responsibility in organizational evaluation processes.

This perspective extends sociotechnical and organizational theories of accountability by locating responsibility not only in evaluative regimes or actor performance but in the material organization of procedural inscription. As AI systems increasingly participate in procedural work, through documentation assistance, decision support, and audit-trail generation, the governance of cognitive friction acquires additional complexity. AI-mediated procedural systems may reduce certain forms of interpretive friction while simultaneously introducing new friction configurations by embedding algorithmic criteria that remain opaque to executing actors. Future work examining

how AI integration reshapes the distribution of cognitive friction in procedural texts would extend the present framework toward the governance of accountability in AI-augmented regulated environments.

This document-level account of responsibility structuring can be situated within a broader anticipatory process of accountability alignment in regulated organizational systems. Prior to action, actors often operate under convergent expectations regarding how their interpretations and executions will later be evaluated, justified, and attributed. AAC conceptualizes this system-level alignment of accountability expectations across procedural, documentary, and evaluative infrastructures. From this perspective, cognitive friction identifies a localized mechanism through which such anticipatory accountability structuring becomes embedded within procedural language itself. Procedural sentences that distribute interpretive resolution to executing actors therefore constitute sites where broader accountability convergence is operationalized at the document level. As illustrated in Figure 4, responsibility production in regulated sociotechnical systems can be understood as a multi-stage process spanning anticipatory alignment, procedural embedding, investigative stabilization, and representational mediation.

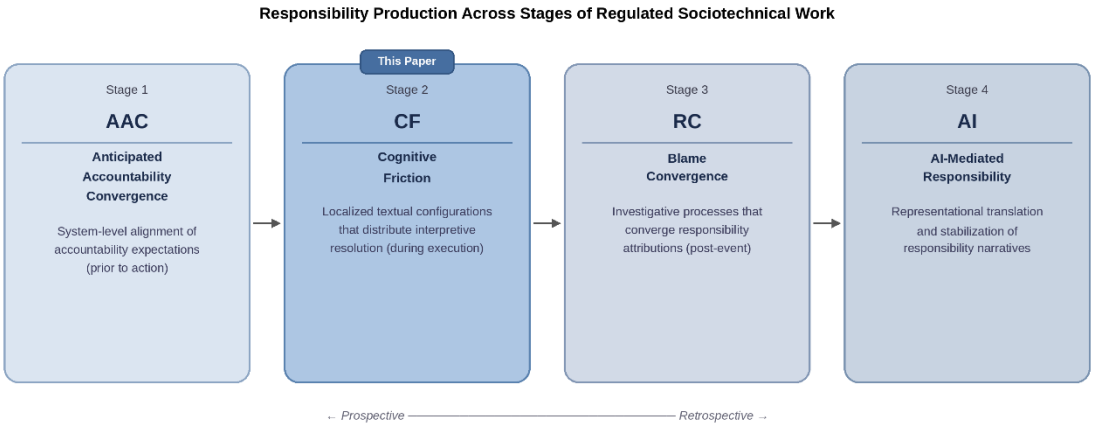

Figure 4. Responsibility production across stages of work in regulated sociotechnical systems. Responsibility is anticipated through system-level alignment of accountability expectations (AAC), embedded within procedural language during execution through cognitive friction, stabilized through investigative attribution processes (blame convergence), and subsequently translated and mediated through representational systems in AI-supported evaluation infrastructures.

AAC operates at the level of system-wide alignment of accountability expectations across procedural and evaluative infrastructures prior to action. Cognitive friction, by contrast, specifies the localized textual configurations through which such anticipatory alignment becomes operationalized during execution. Taken together, these findings suggest that accountability in regulated organizational systems is not only evaluated after action but also materially pre-configured through the design of procedural documentation.

Limitations and Scope Conditions

Several limitations delimit the scope of the present analysis and indicate directions for further empirical development. First, although the empirical corpus draws on multiple forms of publicly available procedural documentation, including laboratory guidance, biosafety protocols, chemical safety procedures, and synthetic SOP-like constructions, it remains restricted to highly formalized procedural genres. Such documents provide analytically clear instances of rule-governed inscription, but cognitive friction as a structural property of procedural language may vary across organizational domains with different documentation traditions, levels of formalization, or accountability regimes. Future research can therefore examine how cognitive friction manifests in other procedural environments, including clinical, engineering, or administrative documentation systems. Cognitive friction thus provides a document-level lens for analyzing how procedural texts structure judgment and responsibility in regulated organizational work.

Second, the present study focuses on sentence-level structural features as minimal units of procedural inscription. This analytical choice enables identification of localized interpretive demands but does not capture higher-order document structures such as section organization, hierarchical nesting of procedures, or cross-document architectures that may also contribute to distributed judgment demands. Extending cognitive friction analysis to multi-sentence or document-level configurations would provide a more comprehensive account of how procedural texts structure interpretive work across organizational documentation systems.

Third, the coding framework identifies structural features associated with interpretive demand but does not measure cognitive effort or decision difficulty directly at the level of actors. Cognitive friction is conceptualized here as a property of procedural text rather than individual cognition; however, linking structural configurations to observed interpretive processes or performance outcomes would strengthen the empirical grounding of the framework. Future studies

combining textual analysis with observational or experimental approaches could examine how structurally embedded judgment demands are experienced and resolved in situated practice.

Finally, the present analysis does not aim to establish causal relationships between procedural design and downstream responsibility attribution outcomes. While the framework specifies how procedural architecture allocates localized interpretive resolution to actors, accountability stabilization remains shaped by organizational evaluation processes, institutional norms, and interactional dynamics beyond the text itself. Cognitive friction therefore captures one upstream component of responsibility structuring within sociotechnical systems rather than a complete explanatory account.

These limitations nonetheless underscore the analytical value of treating procedural language as a site where responsibility-relevant judgment demands become structurally organized. By identifying observable textual configurations that allocate interpretive resolution within rule-governed work, the present framework provides a foundation for broader empirical and comparative investigations of how documentation practices participate in shaping the locus of accountable action across organizational domains.

Future Research Directions

The cognitive friction framework opens several avenues for further empirical and theoretical development. One immediate direction involves extending analysis beyond sentence-level configurations to examine how multi-sentence structures and document architectures distribute interpretive demand across procedural systems. Procedural texts often embed decision criteria through hierarchical nesting, section-level dependencies, and cross-document linkages that cannot be fully captured at the level of isolated sentences. Investigating these higher-order textual structures would enable a more comprehensive account of how documentation organizes judgment across organizational knowledge infrastructures.

A second research direction concerns variation in cognitive friction across procedural domains and institutional contexts. Documentation practices differ substantially across industries in their levels of formalization, regulatory exposure, and accountability regimes. Comparative analysis across laboratory, clinical, engineering, or administrative procedural systems could reveal how domain-specific documentation conventions shape the distribution and intensity of structurally embedded judgment demands. Such comparisons would clarify whether cognitive

friction reflects general properties of procedural language or domain-specific inscription practices.

Third, future work may examine how actors experience and resolve cognitive friction in situated practice. While the present study conceptualizes friction as a structural property of procedural text, its organizational consequences unfold through interaction with users interpreting and enacting procedural instructions. Observational or experimental studies could investigate how practitioners navigate conditional structures, deferred criteria, and cross-references during task execution, and how such navigation shapes decision trajectories and responsibility attribution. Linking textual configurations to observed interpretive processes would deepen the sociotechnical grounding of the framework.

Another promising direction involves integrating cognitive friction with broader theories of organizational routines and documentation infrastructures. Procedural texts form part of distributed organizational memory systems that coordinate action across time and actors. Examining how friction-prone structures interact with routine dynamics, standardization processes, and audit regimes may illuminate how documentation both stabilizes and redistributes judgment within organizations. This perspective would position cognitive friction within ongoing theoretical debates on the performativity of organizational texts and the materialization of accountability structures.

Finally, computational approaches offer opportunities to scale cognitive friction analysis across large document corpora. Automated detection of structural features such as conditionality, cross-referencing, exception handling, and discretionary qualifiers could enable comparative mapping of friction distributions across procedural systems. Such analyses would allow examination of how documentation architectures evolve over time, respond to regulatory change, or vary across organizations. By combining qualitative interpretive analysis with computational text methods, future research could establish cognitive friction as a scalable analytical lens for studying documentation-centered coordination in complex sociotechnical environments.

Conclusion

This study provides initial evidence that responsibility-bearing judgment in regulated organizational work is partly structured within procedural sentence architecture and introduces cognitive friction as a document-level mechanism capturing this structural organization. By analyzing sentence-level configurations of structural features such as conditional branching, cross-

references, exception structures, and discretionary qualifiers, the analysis demonstrates that interpretive demands are not solely products of individual cognition but are partially produced through the architecture of procedural language itself.

Situated within a broader research program on responsibility in regulated sociotechnical systems, cognitive friction specifies how responsibility-bearing judgment becomes structurally embedded within procedural documentation during execution. Procedural texts thereby function not only as instruments of coordination but as infrastructures of accountable action, shaping where interpretive resolution becomes necessary and where responsibility becomes attributable in organizational practice.

By foregrounding the procedural sentence as a minimal unit of inscription, this study identifies a document-level mechanism linking procedural design to the structuring of accountability. Future research may extend this framework across domains and document architectures to examine how variations in procedural configuration influence the organization of judgment and responsibility in complex sociotechnical systems. Cognitive friction thus provides a document-level lens for analyzing how responsibility is materially organized within procedural systems.

Code Availability

The code supporting this study is publicly available at: https://github.com/sorakim-phd/sop-cognitive-friction-pilot

The repository contains the rule-based annotation scripts and analysis workflow used to identify sentence-level cognitive friction features in the 500-sentence procedural corpus analyzed in this study. The implementation operationalizes the structural feature definitions described in the Methods section (conditional branching, cross-references, exception structures, and discretionary qualifiers) and computes composite cognitive friction scores for each sentence.

The repository also includes an illustrative subset of procedural sentences and example annotation outputs to support computational transparency and reproducibility. The implementation is intended as a transparent research prototype demonstrating the computational observability of structurally embedded judgment demands in procedural text, rather than a predictive or production-level system.

Data Availability

The procedural sentence corpus analyzed in this study consists of 500 procedurally realistic sentences derived from publicly accessible regulatory and safety documentation, including WHO laboratory guidance, academic laboratory safety resources, and related procedural materials. Sentences were segmented, selected, and normalized at the procedural statement level for annotation.

Because the corpus includes sentences adapted from publicly available documents and synthetic constructions designed to approximate SOP-like procedural structure, only an illustrative subset and annotation outputs are provided in the public repository to support transparency and reproducibility. The full annotated corpus used in the analysis is available from the author upon reasonable request.

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